MANNESMANN
LINE PIPE
Ein Unternehmen der Salzgitter Gruppe

Inspection certificate 3.1
EN 10204:2004
(A02)

No. (A03) **10001475**
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Date 02.01.2025

No. (A07) Purchaser (A06)	PHP03478 15.10.2024 Salzgitter Mannesmann International (USA) Inc. 1770 St. James Place, Suite 600 77056-3499 HOUSTON USA	No. (A07) Customer (A06)	Industrial Piping Specialists Box 581270 74158 TULSA USA

Product (B01): steel line pipe, HFW-HFI-longitudinally welded

Works order No. (A08): 0603010573
Dispatch note No.: 0080030793
10.12.2024
Inspection (A05): Manufacturer acc.

Steel grade and terms of delivery (B02-B03):
API SPEC 5L 46TH PSL 2 X65M
Hardness acc. to NACE MR0175 / ISO15156
Hardness acc. to NACE MR0103
Hydrostatically tested 100% SMYS
Impact Requirements according to API 5L 46th: Base material at -29°C (M20°F), HAZ at -29°C (M20°F) and Weld seam at -20°C (M4°F) / shear area determination base material for information only.
marking acc. to API 5L 46th

Heat treatment: Weld zone inductive annealed
(Temperature minimum 1600°F / 871°C)

Material data (B01-B99)

Item	Heat No. (B07)	Quantity (B08)	Cond. of delivery (B04)	D x T x Length (B09-B11) in x in x ft	Length (B09-B11) ft	Weight (B12) lbs
040	730213	2	U	16,000 x 0,500 x 44,967	89,783	7.438
040	761641	58	U	16,000 x 0,500 x 44,967	2.598,247	215.254
Σ		60	U: as rolled		2.688,030	222.692

The mentioned length and weight refers to the product in delivery condition

Product description (B06)

Inside conditions (B06)
Surface: black

Outside conditions (B06)
Surface: black temporary anti-corrosive protection laquer approx. 35 my thickness

ends (B06)
End 1: bevelled 30 +5/-0°, root face 1,6 +/-0,8 mm
End 2: bevelled 30 +5/-0°, root face 1,6 +/-0,8 mm
End protection: without plastic caps

Marking of the product (B06)
Longitudinal marking at one end:
Manufacturers sign SIEGEN API SPEC 5L 0035 API monogram
MM-YY 16.000 0.500 X65M PSL 2 HFW
A HEAT NO. Heat-no. Single pipe length (ft) Pipe-no. TESTED 3840 PSI
MADE IN GERMANY IPS P.O. No. PHP03478

Internal marking at one end:
Barcode Pipe-no.
Product No. Manufacturers sign
Single pipe length (ft) HEAT No. Heat-no.
TESTED 3840 PSI IPS

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Ladle analysis (C70–C99)
Manufacturer standard

Heat No. (B07)	C % ≤0,10	Si % ≤0,45	Mn % ≤1,70	P % ≤0,025	S % ≤0,015	Cu % ≤0,50	Cr % ≤0,50	Ni % ≤0,50	Mo % ≤0,50	V %
730213	0,07	0,32	1,22	0,011	0,002	0,02	0,03	0,02	0,003	0,002
761641	0,07	0,32	1,23	0,009	0,002	0,02	0,03	0,02	0,002	0,002

Heat No. (B07)	Ti %	Nb %	B % ≤0,0010	EV1 1) % ≤0,15
730213	0,02	0,04	0,0002	0,06
761641	0,02	0,04	0,0001	0,06

1) EV1 : Nb+V+Ti

Steelmaking process: Basic oxygen process
(C70)

Steel Manufacturer incl. country of melt and pour (A01):
Hüttenwerke Krupp Mannesmann GmbH, Postfach 25 11 24, 47251 DUISBURG, GERMANY

Coil Manufacturer (A01):
ThyssenKrupp Steel Europe AG, Postfach 4 71 61, 47166 DUISBURG, GERMANY

Product analysis (C70–C99)
ISO/TR 9769

Test type: Lot testing

Specimen No. (C00)	Heat No. (B07)	C % ≤0,10	Si % ≤0,45	Mn % ≤1,70	P % ≤0,025	S % ≤0,015	Cu % ≤0,50	Cr % ≤0,50	Ni % ≤0,50
1002975345 *)	730213	0,05	0,33	1,23	0,010	0,001	0,02	0,03	0,02
1002975346 *)	730213	0,05	0,33	1,22	0,009	0,001	0,02	0,03	0,02
1002975278	761641	0,05	0,33	1,24	0,008	0,002	0,02	0,03	0,02
1002975279	761641	0,05	0,33	1,25	0,009	0,002	0,02	0,03	0,02

Specimen No. (C00)	Heat No. (B07)	Mo % ≤0,50	V %	Ti %	Nb %	B % ≤0,0010	EV1 1) % ≤0,25	EV2 2) % ≤0,15
1002975345 *)	730213	0,01	0,002	0,02	0,04	0,0001	0,13	0,07
1002975346 *)	730213	0,01	0,002	0,02	0,04	0,0001	0,12	0,07
1002975278	761641	0,01	0,001	0,02	0,05	0,0001	0,13	0,07
1002975279	761641	0,01	0,001	0,02	0,05	0,0001	0,13	0,07

1) EV1 : PCM = C+Si/30+(Mn+Cu+Cr)/20+Mo/15+Ni/60+V/10+5*B
2) EV2 : Nb+V+Ti
*) The sample pipe is not part of the delivery

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Tensile test (C10–C29)
ISO 6892–1

Test type: Lot testing
Condition specimen: Specimen as delivery condition/Pipe as rolled
Specimen shape: Flat tensile test

Specimen No. (C00)	Heat No. (B07)	Location (C01) 1)	Orient. (C02) 2)	Height of specimen in	Width of specimen in	Yield strength (C11) Rt0,5 psi 65300–87000	Tensile strength (C12) Rm psi 77600–110200	Rt0,5/Rm ≤0,93
1002975278	761641	BM	T180	0,51	1,57	72519	89488	0,81
1002975345 *)	730213	BM	T180	0,51	1,57	76145	93114	0,82

Specimen No. (C00)	Heat No. (B07)	Location (C01) 1)	Orient. (C02) 2)	Elongation (C13) 2in % ≥23
1002975278	761641	BM	T180	37
1002975345 *)	730213	BM	T180	39

Specimen No. (C00)	Heat No. (B07)	Location (C01) 1)	Orient. (C02) 2)	Height of specimen in	Width of specimen in	Tensile strength (C12) Rm psi 77600–110200
1002975278	761641	WL	T000	0,51	1,57	81511
1002975345 *)	730213	WL	T000	0,51	1,57	81946

1) BM: Base material
WL: Weld seam
2) T000: Transversal 0°
T180: Transversal 180°
*) The sample pipe is not part of the delivery

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Charpy impact test (C40–C49)
ASTM A 370

Test type: Lot testing
Condition specimen: Specimen as delivery condition/Pipe as rolled
Type of notch: V–notch
Specimen size: 1/1 specimen

Specimen No. (C00)	Heat No. (B07)	Location (C01) 1)	Orient. (C02) 2)	Temp. (C03) °F	Impact energy (C42–C43) 1 ftlbf ≥15	2	3	MW 3) ftlbf ≥20
1002975278	761641	BM	T090	−020	151	155	150	152
1002975345 *)	730213	BM	T090	−020	173	169	170	171

Specimen No. (C00)	Heat No. (B07)	Location (C01) 1)	Orient. (C02) 2)	Temp. (C03) °F	Shear fracture (C42–C43) 1 %	2	3	MW 3) %
1002975278	761641	BM	T090	−020	90	90	90	90
1002975345 *)	730213	BM	T090	−020	90	90	90	90

Specimen No. (C00)	Heat No. (B07)	Location (C01) 1)	Orient. (C02) 2)	Temp. (C03) °F	Impact energy (C42–C43) 1 ftlbf ≥15	2	3	MW 3) ftlbf ≥20
1002975278	761641	HAZ	T000	−020	181	185	185	184
1002975278	761641	WL	T000	−004	177	165	162	168
1002975345 *)	730213	HAZ	T000	−020	181	184	181	182
1002975345 *)	730213	WL	T000	−004	218	168	219	202

1) BM: Base material
HAZ: Heat affected zone
WL: Weld seam
*) The sample pipe is not part of the delivery

2) T000: Transversal 0°
T090: Transversal 90°
3) MW: Average

Hardness test (C30–C39)
ISO 6507–1

Test type: Lot testing

Specimen No. (C00)	Heat No. (B07)	Location 1)	Härtetyp 2)	Hardness max. HV10 ≤250				
1002975278	761641		HV	218				
1002975345 *)	730213		HV	226				

2) HV: hardness Vickers
*) The sample pipe is not part of the delivery

Destructive tests (D51–D99)

Flattening test according to ISO 8492:	Satisfactory
Metallographic determination of micro and macro structure:	Satisfactory

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Non–destructive tests (D02–D50)

Visual and dimensional inspection:	Satisfactory
Ultrasonic testing of weld seam for the detection of longitudinal imperfections according to ISO 10893–11, acceptance level U3:	Satisfactory
Hydrostatic test 3840 PSI, holding time 5 s:	Satisfactory
Check for residual magnetism at the pipe ends max. 30 Gauss:	Satisfactory

Pipe list (B01–B99)

Item	Pipe No. (B07)	Product No. (B07)	Heat No. (B07)	Inspection No. (B07)	Cond. of delivery (B04)	Length ft (B09–B11)	Weight lbs (B12)
040	1002975274	25450101	761641	0	U	44,885	3.719
040	1002975275	25450102	761641	0	U	44,885	3.719
040	1002975276	25450103	761641	0	U	44,898	3.719
040	1002975277	25450104	761641	0	U	44,885	3.719
040	1002975278	25450105	761641	0	U	44,869	3.717
040	1002975279	25450106	761641	0	U	44,895	3.719
040	1002975280	25450107	761641	0	U	44,888	3.719
040	1002975281	25450108	761641	0	U	44,895	3.719
040	1002975282	25450109	761641	0	U	44,892	3.719
040	1002975283	25450110	761641	0	U	44,892	3.719
040	1002975284	25450111	761641	0	U	44,902	3.721
040	1002975285	25450112	761641	0	U	44,895	3.719
040	1002975286	25450113	761641	0	U	44,895	3.719
040	1002975288	25450201	761641	0	U	44,892	3.719
040	1002975289	25450202	761641	0	U	44,892	3.719
040	1002975290	25450203	761641	0	U	44,888	3.719
040	1002975291	25450204	761641	0	U	44,898	3.719
040	1002975292	25450205	761641	0	U	44,895	3.719
040	1002975293	25450206	761641	0	U	44,895	3.719
040	1002975294	25450207	761641	0	U	44,895	3.719
040	1002975295	25450208	761641	0	U	44,898	3.719
040	1002975296	25450209	761641	0	U	44,898	3.719
040	1002975297	25450210	761641	0	U	44,892	3.719
040	1002975298	25450211	761641	0	U	44,895	3.719
040	1002975299	25450212	761641	0	U	44,895	3.719
040	1002975300	25450213	761641	0	U	44,892	3.719
040	1002975302	25450301	761641	0	U	44,888	3.719
040	1002975303	25450302	761641	0	U	44,895	3.719
040	1002975304	25450303	761641	0	U	44,885	3.719
040	1002975305	25450304	761641	0	U	44,885	3.719
040	1002975306	25450305	761641	0	U	44,895	3.719
040	1002975307	25450306	761641	0	U	44,888	3.719
040	1002975308	25450307	761641	0	U	44,898	3.719
040	1002975309	25450308	761641	0	U	44,888	3.719
040	1002975310	25450309	761641	0	U	44,902	3.721
040	1002975312	25450401	761641	0	U	44,885	3.719
040	1002975313	25450402	761641	0	U	44,888	3.719
040	1002975314	25450403	761641	0	U	44,895	3.719
040	1002975315	25450404	761641	0	U	44,888	3.719
040	1002975316	25450405	761641	0	U	44,898	3.719
040	1002975317	25450406	761641	0	U	44,882	3.719
040	1002975319	25450408	761641	0	U	44,885	3.719
040	1002975320	25450409	761641	0	U	44,888	3.719
040	1002975321	25450410	761641	0	U	44,895	3.719
040	1002975322	25450411	761641	0	U	44,892	3.719
040	1002975323	25450412	761641	0	U	44,902	3.721

U : as rolled

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Pipe list (B01–B99)

Item	Pipe No. (B07)	Product No. (B07)	Heat No. (B07)	Inspection No. (B07)	Cond. of delivery (B04)	Length ft (B09–B11)	Weight lbs (B12)
040	1002975324	25450413	761641	0	U	44,892	3.719
040	1002975326	25450501	761641	0	U	44,892	3.719
040	1002975327	25450502	761641	0	U	44,895	3.719
040	1002975328	25450503	761641	0	U	44,885	3.719
040	1002975329	25450504	761641	0	U	44,902	3.721
040	1002975330	25450505	761641	0	U	44,882	3.719
040	1002975333	25450508	761641	0	U	39,406	3.265
040	1002975334	25450509	761641	0	U	44,888	3.719
040	1002975335	25450510	761641	0	U	44,898	3.719
040	1002975336	25450511	761641	0	U	44,898	3.719
040	1002975337	25450512	761641	0	U	44,888	3.719
040	1002975338	25450513	761641	0	U	44,898	3.719
040	1002975340	25450601	730213	0	U	44,888	3.719
040	1002975351	25450612	730213	0	U	44,895	3.719
Quantity (B08) 60		U: as rolled				2.688,030	222.692

Statement of compliance:
The products have been tested in accordance with the purchase specification and are found to be satisfactory. Alteration of this testimonial and certification or use for other products is not allowed and are regarded as falsification of documents and will be subjected to criminal jurisdiction.

We hereby certify that the delivered material complies with the terms of the order.
(Z01)

Trademark
(A04)

Mannesmann Line Pipe GmbH
In der Steinwiese 31
57074 SIEGEN
GERMANY
(A01)

A Inspection Stamp (Z03)

Inspection Representative (Z02)
Sascha Arndt

Production Bend Report

Tulsa Tube Bending

Customer:	DNOW	**Report No.:**	109710-PBR-1
Customer PO No.:	663013318	**Code of Fabrication:**	
TTB Job No.:	26H109710	B16.49-2023/42414SP	

Mother Pipe & Bend Information

OD (in): 16.000 Grade: API-5L-X65 PSL 2 Radius (in): 80 Bending Machine ID: M2 Heat No.: 761641

Essential Variables

Wall Thickness (inch): 0.500 Radius-to-Diameter Ratio: 5.0 Coil Design: TTB High Spec Coolant Type: Water

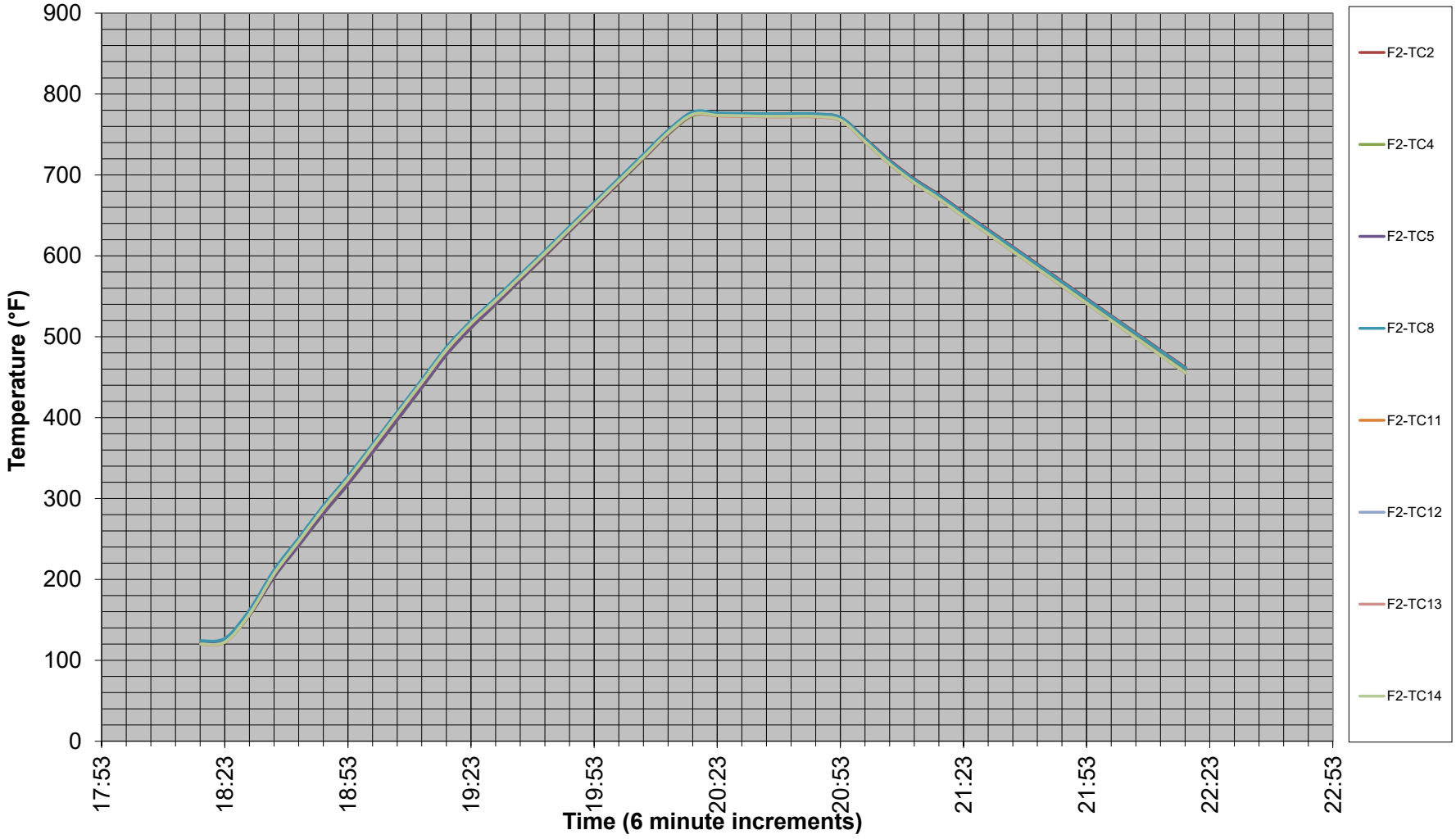
Weld Seam Location: 0 degrees Heat Treatment Holding Temp (°F): 775 Heat Treatment Soak Time (min): 30.0

Note 1: Bending recipe (Bend Speed & Bend Temp parameters) are proprietary to Tulsa Tube Bending Co. and can be made available to the purchaser or the purchaser's third party inspector(s) at the bend facility.

Bend ID No.	Operator Initials	Date	Bending Speed (ipm)	Extrados Bend Temp (°F):	Intrados Bend Temp (°F):	Coolant Temp (°F)	Coolant Flow Rate (GPM)	Induction Frequency (Hz)
Qualification Bend Value:			See Note 1.			80.4 - 82.8	11.1	803
Limits of Variation:			+/- 0.1 ipm	+/- 50°F	+/- 50°F	+/- 25°F	+/- 10%	+/- 20%
109710-1-1	JC	7/30/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	84.02 - 86.54	11.4	804
109710-1-2	JC	7/30/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	84.38 - 92.66	11.4	802
109710-1-3	JC	7/30/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	84.2 - 87.8	11.4	803
109710-1-4	JC	7/30/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	84.38 - 85.28	11.5	803
109710-2-1	JC	7/30/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	86.9 - 89.78	11.4	805
109710-2-2	JC	7/31/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	79.34 - 88.88	11.2	804
109710-2-3	JC	7/31/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	81.5 - 85.28	11.7	805
109710-2-4	JC	7/31/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	82.76 - 85.28	11.4	805
109710-3-1	JC	7/30/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	84.74 - 87.62	11.4	805
109710-3-2	JC	7/30/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	84.56 - 93.38	11.2	806
109710-3-3	JC	7/30/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	84.56 - 93.38	11.2	806
109710-3-4	JC	7/31/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	81.14 - 83.48	11.3	806
109710-3-5	JC	7/31/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	82.4 - 84.56	11.6	806
109710-3-6	JC	7/31/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	83.12 - 86.18	11.8	806
109710-3-7	JC	7/31/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	83.66 - 85.82	11.1	806
109710-3-8	JC	7/31/2026	+/- 0.1 ipm	+/- 50°F	+/- 50°F	84.56 - 87.26	11.6	805

Customer	**Tulsa Tube Bending Company**	**Zentech Job 1-30-26-0021096**	**8/5/2026**
Job	**26H109710-1**		
PO	**26H109710-1**		
Part No.	**Multi**		

Zentech Thermal Processing
Furnace No. F-2



Calibration Date:	**7/1/2026**			**Zentech Operator**
Calibration Due:	**10/1/2026**	**Thermocouple Type :**	**K**	**Signature :**

Production Bend Hardness Report

Tulsa Tube Bending

Customer:	DNOW	**Report No.:**	109710-H-1
Customer PO No.:	663013318	**Code of Fabrication:**	
TTB Job No.:	26H109710	B16.49-2023/42414SP	

Mother Pipe Information

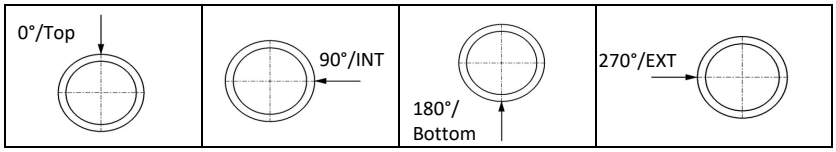
Pipe Size (in): 16 x 0.500 Grade: API-5L-X65 PSL 2

Process Information

Instrument Model No.: UCI-3000H Probe Serial No.: MK8823417972

Serial No.: MK8821415112-1754

Key:



T= Top (Weld if Welded Material)
H= Weld Heat Affected Zone
I= Intrados of Bend
E= Extrados of Bend
TAN= Tangent of Bend
STZ= Start Transition Zone
FTZ= Finish Transition Zone
3/4= 3/4 Point of the Bend Arc

Note 1. Weld & Weld HAZ readings are only required on bends made from welded mother pipe.

Note 2. Readings at 3/4 bend arc locations are only required for production bends at least 30° greater than the QB angle.

Hardness Readings

Min Allowed Values:	156	Max Allowed Values:	238

Bend ID No.	TAN T 0°	TAN H 0°	TAN 270°	STZ E 270°	STZ I 90°	MID T 0°	MID H 0°	MID E 270°	MID I 90°	FTZ E 270°	FTZ I 90°	3/4 T 0°	3/4 H 0°	3/4 EXT 270°	3/4 INT 90°
109710-1-1	192	186	199	195	190	196	189	193	187	189	189	N/A	N/A	N/A	N/A
109710-1-2	201	193	198	193	195	192	190	191	191	195	187	N/A	N/A	N/A	N/A
109710-1-3	190	188	192	195	191	202	195	189	201	190	190	N/A	N/A	N/A	N/A
109710-1-4	207	198	191	193	196	201	197	189	195	193	193	N/A	N/A	N/A	N/A
109710-2-1	202	191	207	193	191	198	191	196	191	196	194	N/A	N/A	N/A	N/A
109710-2-2	199	194	198	195	194	198	195	190	190	198	192	N/A	N/A	N/A	N/A
109710-2-3	212	197	194	197	195	205	198	193	198	193	196	N/A	N/A	N/A	N/A
109710-2-4	196	194	191	187	186	198	195	194	197	191	190	N/A	N/A	N/A	N/A
109710-3-1	198	192	201	197	196	198	193	197	193	199	193	N/A	N/A	N/A	N/A
109710-3-2	201	198	199	200	189	197	192	197	191	200	197	N/A	N/A	N/A	N/A
109710-3-3	203	199	192	199	190	202	195	197	186	197	187	N/A	N/A	N/A	N/A
109710-3-4	201	198	192	193	199	195	189	193	187	195	197	N/A	N/A	N/A	N/A
109710-3-5	191	187	194	196	198	193	190	190	195	192	193	N/A	N/A	N/A	N/A
109710-3-6	196	190	197	194	198	195	189	188	196	191	195	N/A	N/A	N/A	N/A
109710-3-7	197	190	195	197	191	194	193	197	189	191	196	N/A	N/A	N/A	N/A
109710-3-8	200	197	195	190	192	194	187	196	195	197	196	N/A	N/A	N/A	N/A

READINGS: ☑ MEET ACCEPTANCE CRITERIA ☐ DO NOT MEET ACCEPTANCE CRITERIA

PERFORMED BY: JV

DATE: 8/13/2026

Dimensional-Visual Inspection Report

Tulsa Tube Bending

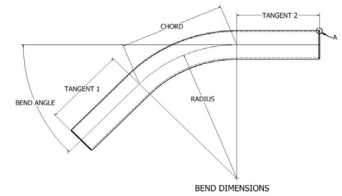
Customer:	DNOW	**Pipe Size (in):**	16 x 0.500	**Report No.:**	109710-DV-1-1
Customer PO No.:	663013318	**Grade:**	API-5L-X65 PSL 2	**Bend ID No.:**	109710-1-1
TTB Job No.:	26H109710	**Heat No.:**	761641	**Code:**	B16.49-2023/42414SP

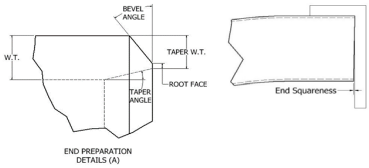
Physical Dimensions

Dimension	Target	Tolerance	Actual
Tangent 1 (in)	24.0	+/- 0.19 in	24.00
Tangent 2 (in)	24.0	+/- 0.19 in	24.13
Chord (in)	13.9	n/a	13.9
Angle (deg)	10.0	+/- 0.5 °	10.0
Radius, R (in)	80	+/- 1%	80.0
Performed By:	DW	**Date:**	8/17/2026

End Prep Dimensions

Dimension	Target	Tolerance	Actual	
Bevel Angle (deg)	32.5	+/- 2.5°	32.50	END 1
Root Face (in)	0.06	+/- 0.03 in	0.06	
Angle (deg)	n/a	n/a	n/a	
Min ID	15	+/- 0.1 in	14.94	
Max ID	15	+/- 0.1 in	15.00	
Max End Sq. (in)	0	+/- 0.09 in	0.07	
Bevel Angle (deg)	32.5	+/- 2.5°	32.50	END 2
Root Face (in)	0.06	+/- 0.03 in	0.06	
Taper Angle (deg)	n/a	n/a	n/a	
Min ID	15	+/- 0.1 in	14.94	
Max ID	15	+/- 0.1 in	14.94	
Max End Sq. (in)	0	+/- 0.09 in	0.07	
Performed By:	DW	**Date:**	8/17/2026	





Ovality **[(O.D. max – O.D. min)/O.D. nominal] x 100**

Quadrant	Limit	Min OD	Max OD	% Ovality
End 1	1.0%	15.952	15.976	0.2%
Start of Bend	3.0%	15.943	15.966	0.1%
1/4 of Bend Arc	3.0%	15.956	15.981	0.2%
1/2 of Bend Arc	3.0%	15.987	16.017	0.2%
3/4 of Bend Arc	3.0%	15.953	16.011	0.4%
Stop of Bend	3.0%	15.989	16.001	0.1%
End 2	1.0%	15.975	15.980	0.0%
Performed By:	DW		**Date:**	8/17/2026

Wall Thinning of Extrados

Quadrant	Limit	Nominal Wall Thickness	Extrados Wall Thickness	% Wall Thinning
Start of Bend	10.0%	0.500	0.477	4.6%
1/4 of Bend Arc	10.0%	0.500	0.472	5.6%
1/2 of Bend Arc	10.0%	0.500	0.475	5.0%
3/4 of Bend Arc	10.0%	0.500	0.474	5.2%
Stop of Bend	10.0%	0.500	0.482	3.6%
Performed By:	DW		**Date:**	8/17/2026

ID Gauge Drift/Pig Inspection

Minimum Gauge Diameter =	97.0%	of Nominal ID, or	14.550 in
Gauge Length Required =	15 in		

Gauge passes through the entire bend without power equipment assistance

Performed By:	DW	**Date:**	8/17/2026

Visual Inspection

All accessible surfaces found to be free of visible injurious defects.
Bend is stenciled with correct information.

Performed By:	DW	**Date:**	8/17/2026

READINGS: ✓ MEET ACCEPTANCE CRITERIA

☐ DO NOT MEET ACCEPTANCE CRITERIA

GLOBE X-RAY SERVICES

8441 South Union Tulsa, OK. 74132

Certification: 197451-1

Report Of Ultrasonic Examination

Customer: Tulsa Tube Bending Job # 26H109710-1

Address:

Description:

The items listed below have been ultrasonically examined and meet the requirements in accordance with the attached calibration.

Unless otherwise noted by an * following the joint number, there were no recordable indications observed.

Description	Joint No.	Accept	Reject
Long Seam	109710-1-1,109710-1-2,109710-1-3,109710-1-4	√	
Long Seam	109710-2-1,109710-2-2,109710-2-3,109710-2-4	√	
Long Seam	10710-3-1,10710-3-2,10710-3-3,10710-3-4	√	
Long Seam	10710-3-5,10710-3-6,10710-3-7,10710-3-8	√	

Examination By: Colter Moyer Qualification Level : II

Printed: Colter Moyer Date : August 13, 2026

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GLOBE X-RAY SERVICES

8441 South Union Tulsa, OK. 74132

Certification: 197451-1

UT Examination Record - Calibration Data Report

Customer: Tulsa Tube Bending Date: August 13, 2026

Job # 26H109710-1 Reference Code/Specification : ASME sec. V, API5L

Procedure # GXS-II-F Rev No. 1-26 Acceptance Standard: ASME sec. VIII, API5L

Instrument: Olympus Epoch 650 S/N: 231803608 Couplant : x Sonix ___ Other: ___

Cable Type: CO-Axial RG-174 x 6' Long

Method:

[X] Contact [X] Manual Scan [X] Pulse Echo [X] Single Element [ ] Dual Element [ ] Array

General Calibration (Systems Cal.) Prior to Examination

Search Unit							Instrument Settings							
Search Unit	Beam Angle	Size	Frequency MHz	S/N	Wave Form Shear	Wave Form Longitude.	Damping	Range	Delay	Reject	Frequency MHz	Init.Cal Check	Cal Check	Final Cal Check
#1	0°	1"	2.25 MHz	U12708		X	50	2.5	1.07	0	2.25 MHz	1230		1300
#2	70°	1/2"	2.25 MHz	SE0679	X		400	5	9.27	0	2.25 MHz	1300		1400
#3	45°	1/2"	2.25 MHz	SE0646	X		400	5	10.88	0	2.25 MHz	1400		1500
#4														

Calibration Block Identity

Unless noted, Beam Angle, Resolution, Beam Exit Point, and Sound Path Calibration were Determined on an IIW Block Type II

Serial # 94-5907 Material Type P-1

Type	Serial Number	Material Type	Temp	Reflector	Ref. Gain	Ref. Ind. Amplitude	DAC When Used Electric	DAC When Used Manual	DAC Range When Used	Search Unit
ASME Curve	P1-14/80-1	P-1	Amb.	3/32" SDH	57.5 Db	80 %FSH	X		2.92 S.P.	#2
ASME Curve	P1-14/80-1	P-1	Amb.	3/32" SDH	56.6 Db	80 %FSH			1.41 S.P.	#3

Examination By: Colter Moyer Qualification Level : II

Printed: Colter Moyer Date : August 13, 2026

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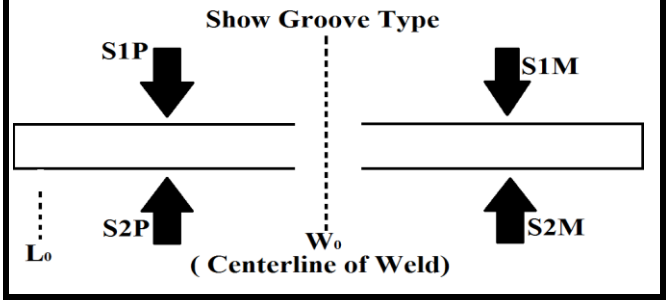
GLOBE X-RAY SERVICES

8441 South Union Tulsa, OK. 74132

Longitudinal Welds Legend

Exam	Scan Number	Type of Scan	Surface Number
X	1	Straight Beam Scan – lamination exam prior to weld exam	S1
X	2	Angle Beam Scan Perpendicular to the Weld from Surface	S-1P
X	3	Angle Beam Scan Perpendicular to the Weld from Surface	S-1M
	4	Angle Beam Scan Perpendicular to the Weld from Surface	S-2P
	5	Angle Beam Scan Perpendicular to the Weld from Surface	S-2M
X	6	Angle Beam Scan Parallel to the Weld from the Surface	S-1P
X	7	Angle Beam Scan Parallel to the Weld from the Surface	S-1M
	8	Angle Beam Scan Parallel to the Weld from the Surface	S-2P
	9	Angle Beam Scan Parallel to the Weld from the Surface	S-2M
	10	Straight Beam Scan- Weld Exam, Surface	S-1P & S-1M, Weld Ground Flush
	11	Straight Beam Scan- Weld Exam, Surface	S-2P & s-2M, Weld Ground Flush

Transducer Coordinates	
L_0	0° or 0 inch of Nozzle Layout (Starting Point for Weld Layout)
L_1	Distance from L_0, in Clockwise Direction to Centerline of Transducer at 25% signal amplitude, near end of Indication.
L_{Max}	Distance from L_0, in Clockwise Direction to Centerline of Transducer at Max Signal Amplitude of Indication.
L_2	Distance from L_0, in Clockwise Direction to centerline of transducer at 25% signal amplitude, at far end of Indication.
W_0	Centerline of Weld
W_{Max}	Distance from W_0, to Exit Point of Transducer at Max Signal Amplitude of Indication.
SP	Sound Path Distance to Indication



Scanning Surface Identification:

S1: Vessel Outside Surface

S2: Vessel Inside Surface

P: Plus side of reference point W_0 ; Surface on the clockwise direction of the 0 Deg point as viewed from vessel top for vertical, vessel as viewed from the inlet/reference end for horizontal vessel.

M: Minus side of reference point W_0 ; Surface on the counter-clockwise direction of the 0 point as viewed from vessel top for vert-vessel as viewed from the inlet/reference end for horizontal vessel.

Starting Point for Weld Layout

Longitudinal Weld:
- [ ] Vertical Vessel: Centerline of circumferential weld at top of longitudinal weld
- [X] Horizontal Vessel: Centerline of circumferential weld on inlet/reference end

Layout Increment Spacing: _____ Number of Layout Points: _____

Weld Accessibility:
- [X] 100% Accessible
- [ ] Restricted Access - Weld Examined with Angle Beam from one side of weld axis only.
- [ ] Inaccessible area(s)- Area of weld which cannot be examined from at least one side with angle beam.
- [ ] Inaccessible from increment number _____ to _____ / _____ to _____
  _____ to _____ / _____ to _____

Examination By: Colter Moyer Colter Moyer Qualification Level : II

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GLOBE X-RAY SERVICES

8441 South Union Tulsa, OK. 74132

Certification: 197451-1

UT Examination Record

Customer: Tulsa Tube Bending Job# 26H109710-1

Weld Identification: See cover page

Material Type: C.S. Material Thickness: .500"

Material Temperature: Ambient Material Diameter: 16"

Test Conditions

Surface Condition of Weld: ____ Ground Flush ____ Ground Smooth X As Welded

Surface Condition of the Scan Surface: X Buffed Smooth ____ Other: ____

Results of Ultrasonic Examination

Accept

No recordable indications noted.

See Attached Legend and Calibration record for the following data.

Examination Scan Numbers this Examination: 1,2,3,6,7

Indication #	Scan No.	Search Unit No.	Beam Angle	% DAC	Search Unit Location		Length Dimensions		Reflector Depth	Reflector Length	Sound Path	Eval.	Results
					W Max	L Max	L1	L2					

Examination By: Colter Moyer Qualification Level : II

Printed: Colter Moyer Date : August 13, 2026

GLOBE X-RAY ASSUMES NO RESPONSIBILITY FOR THE INTEGRITY OF THE STRUCTURE OR FOR ANY LOSSES OF ANY KIND DUE TO ANY KIND OF INTERPRETATION

Globe X-Ray Services, Inc.
8441 S. Union Ave.
Tulsa, OK 74132
918-446-1696

Certification # 195947

Magnetic Particle Examination Record

Customer: Tulsa Tube Bending Date: 8/13/2026

Job/P.O. No.: 109710 Reference Code/Specification: ASME SEC V

MT Procedure No. GXS IV - C Rev. No. 1/26 Acceptances: ASME SEC VIII Div. I

Magnetizing Equipment: Parker Contour Probe B300 Serial No. 1861 Technique: X Yoke _____ Prods

Current Type: X A/C _____ D/C Magnetic Particle Type: _____ Wet X Dry _____ Fluorescent X Visible

Amperage (Prod Technique Only) _____ Amps/inch of Prod Spacing.

Part/Item Material Type: Carbon Material Thickness: Varies

Lighting Equipment: _____ Fluorescent Light Model # _____ Serial # _____
X White Light using Parker Research Y300 or Y400 Probe light
_____ Natural White Light
_____ Other: _____
Light Intensity on surface to be examined: 100 Min foot candles N/A µW/cm²

Comments: Test The Entire Extrados (Back Half) of the Bend. From N.A to N,A, Including Weld Seam
Verify Residual Magnetism at Each End is <25 Gauss

Examination Results				
Qty:	Part Identification	Accept	Reject	Indication Description
16	16" O.D. X 0.5" Wall, 80" CLR, API-5L-X65 PSL 2 ERW Pipe HT# 761641 Bend ID # 109710-1-1, 109710-1-2, 109710-1-3, 109710-1-4, 109710-2-1, 109710-2-2, 109710-2-3, 109710-2-4, 109710-3-1, 109710-3-2, 109710-3-3, 109710-3-4, 109710-3-5, 109710-3-6, 109710-3-7, 109710-3-8	X		None

Examine: Jaime Vera Qualification Level: II

GLOBE X-RAY ASSUMES NO RESPONSIBILITY FOR THE INTEGRITY OF THE STRUCTURE OR FOR ANY LOSSES OF ANY KIND DUE TO ANY KIND OF INTERPRETATION.

Tulsa Tube Bending
4192 S. Galveston Ave.
Tulsa, OK 74107
Tel: 918-446-4461
Fax: 918-446-1595
Serving others. Building people. Pursuing excellence.

Certificate of Conformance

Customer:	DNOW	Certificate No.:	20260818
Customer PO No.:	663013318	Issue Date:	8/18/2026
TTB Job No.:	26H109710		
Drawing No.:	n/a	Revision:	0
Material Size, Grade:	16.00 inch OD x 0.500 inch w.t., API-5L-X65 PSL 2 ERW	Quantity of Bends:	16
Material Heat No(s):	761641		

Bend ID Nos.:	109710-1-1	109710-1-2	109710-1-3	109710-1-4	109710-2-1
	109710-2-2	109710-2-3	109710-2-4	109710-3-1	109710-3-2
	109710-3-3	109710-3-4	109710-3-5	109710-3-6	109710-3-7

We hereby certify that the aforementioned induction bends have been inspected and conform to all of the requirements of the Purchase Order. The induction bends have been manufactured in conformance with:

B16.49-2023/42414SP

*With the following notes and exceptions:

(1) TTB is using welded HFW pipe instead of the SMLS pipe required by spec

(2) TTB will do charpy testing at +14F as per ASME B16.49 and not to -20F as required by the spec.

(3) TTB will only do UT of the weld seam (if welded HFW pipe is approved) and not RT as required by the spec.

(4) External Coating NOTES & EXCEPTIONS to Consumers Energy Coating Spec 26907sp per TTB Quote No. 269641.

Approved By: Austin L Carter Date: 8/18/2026

Keith INDUSTRIES
20701 FM 521 Rosharon , TX 77583
+1-713-991-5670

Coating Inspection Record

KF-010 FBE 5/27/22 REV2

Customer: TTB	Date: 9/2/2026	Items: 12 PCS
Customer PO: 26H 109710-1	Paint Spec: CONSUMER ENERGY 26907SP	

Surface Preparation

Item Salt Test	N/A☐	Result: <1	SSPC SP-1 IF NEEDED		Date	9/2/2026	9/2/2026
Abrasive Salt Test	N/A☐	Result: <1	Cleanliness Required	SSPC SP-10	Time	8:30AM	10:30AM
Laboratory Testing	☐YES ☒No		Cleanliness Achieved	☒Yes☐NO	Air Temp	83°F	88°F
Type of Abrasive	Steel Grit		Profile Required	2.0-4.0 mils	Relative Humidity	78%	76%
Nozzle PSI	☐ N/A 100 psi ☒		Profile Achieved	☒Yes☐NO	Steel	84°F	94°F
Blotter Test	☒PASS		SSPC-VIS 1 Used	☒Yes ☐No	Dew Point	75°F	80°F
Dust Free Surfaces	☒YES ☐No		Visual Inspection	☒Yes ☐No	Stage of Operation	Pre-Blast	Pre-Coating

Coating Application

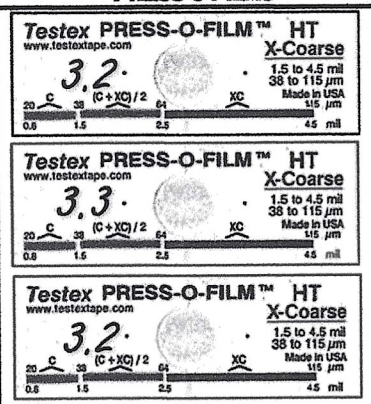
Coat 1st	Product Name: PipeClad 2000	Batch: CH1876UE	Min Mils: 14	Target Pre Coat Temp: 450 deg F

Final Testing

HVST @125v/Mil; 2000v MIN PASS ☒	Adhesion Knife Test: PASS ☒	DFT Test: PASS ☒	All Requirements Met: PASS ☒

KI#	Description	Mil Average
1-1	16" X .500 10 DEG BEND	22
1-2	16" X .500 10 DEG BEND	19
1-3	16" X .500 10 DEG BEND	21
1-4	16" X .500 10 DEG BEND	23
3-1	16" X .500 45 DEG BEND	24

Press O Films



KI#	Description	Mil Average
3-2	16" X .500 45 DEG BEND	26
3-3	16" X .500 45 DEG BEND	20
3-4	16" X .500 45 DEG BEND	31
3-5	16" X .500 45 DEG BEND	25
3-6	16" X .500 45 DEG BEND	36

Verified by Keith: